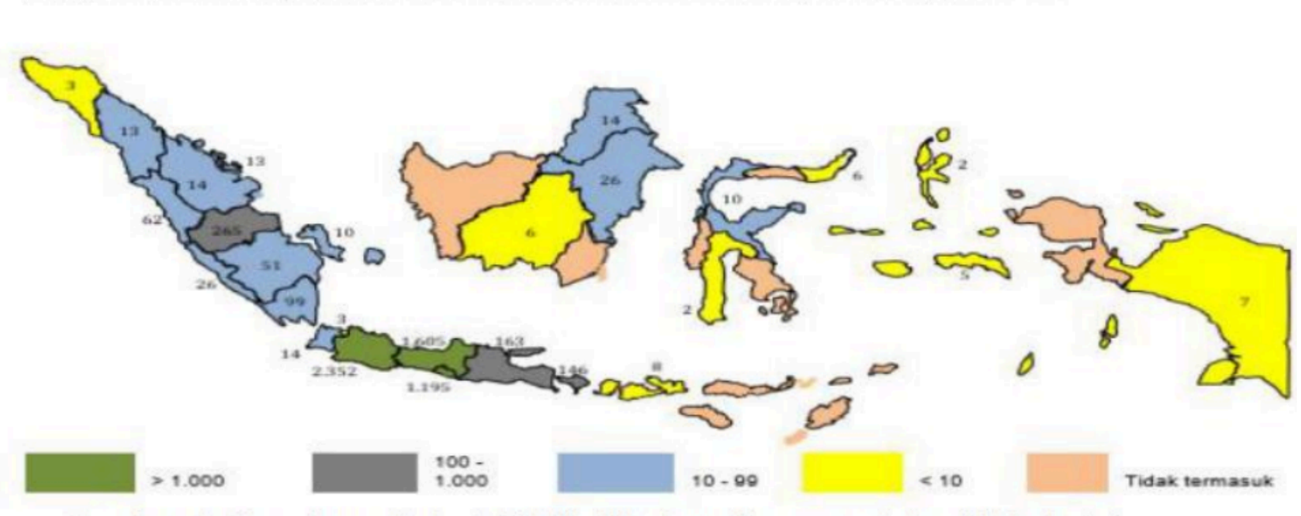
5,849

Distinct batik motifs,
each with a unique story.

Only 10%

Motifs are widely known
and appreciated



*Research by the Bandung Fe Institute and Sobat Budaya in 2015

78%

Young Indonesians aged 18-25 expressed
a desire to learn more about their cultural heritage.

Only 30%

Actively engage in cultural activities,
such as batik-making workshops or heritage tours.



Survey by the Indonesian Youth Development Index

Key Problem

Batik Variants

Indonesia's diverse batik styles are both a strength and a challenge.

Youth Awareness

Young people often know little about batik due to poor education, global fashion influence, and the view that batik is old-fashioned.

Preserving Batik

We need to actively teach and promote batik to ensure it's valued by future generations.



C241-PS409

Team Members



Yohanes
Machine Learning



Adam Haidar Azizi
Machine Learning



Arafat Dhiya 'Ulhaq
Machine Learning



M. Armand Giovani
Mobile Development



Hanafi Satriyo Utomo S.
Mobile Development



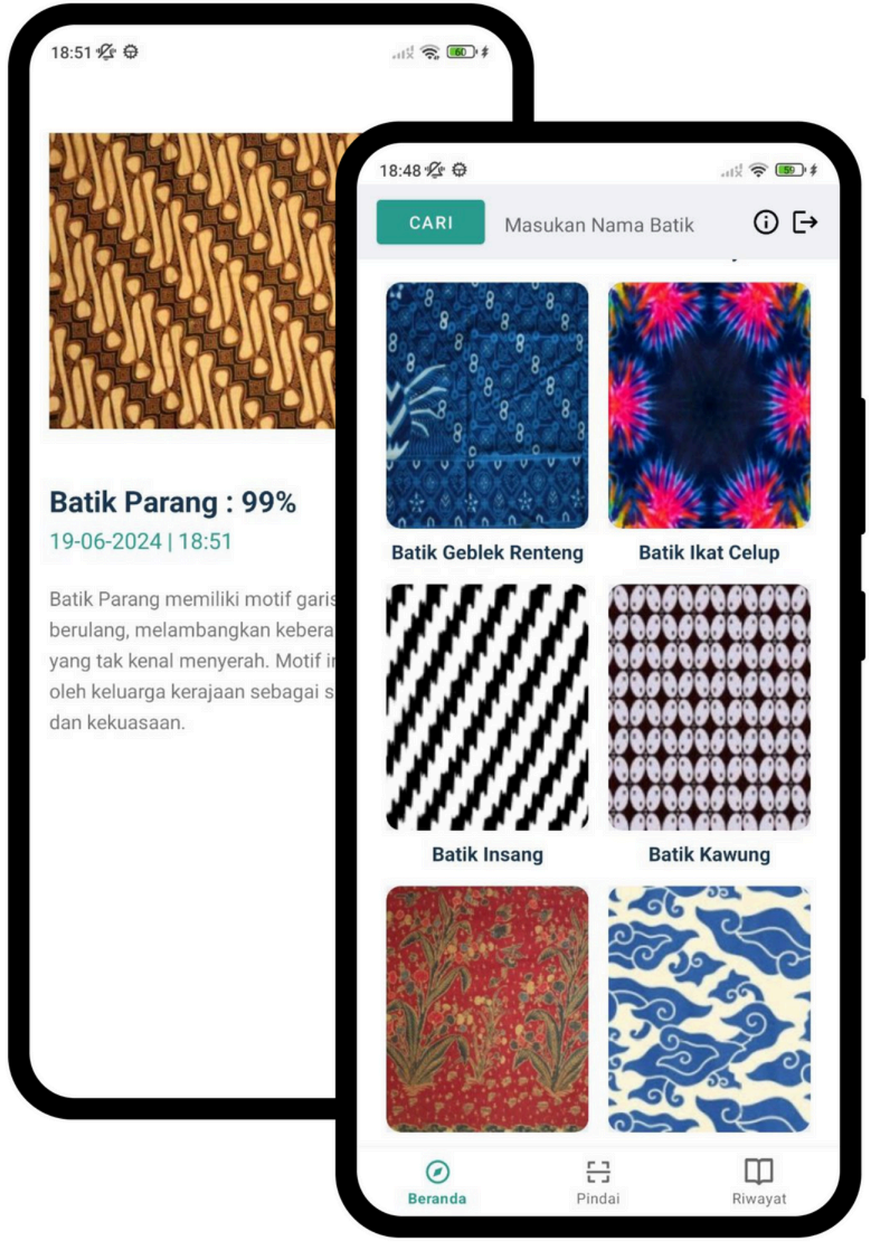
Lutfi Farhan Prayoga
Cloud Computing



Daffa Farhan Raharja
Cloud Computing

Batik Hub

BatikHub is a mobile app that uses machine learning to identify different batik motifs from images. It also educates younger generations about the cultural heritage of batik.



Predicts

Predict the Motif Name of the Batik and from uploaded images based on image batik motifs classification from our dataset

Insight

Provides some insights regarding the Motifs from definition, uniqueness, and philosophies

MODEL SUMMARY

Machine Learning

- MobileNetV2 Model. Utilizes MobileNetV2 as the base, pretrained on ImageNet. Adds GlobalAveragePooling2D, Dropout (0.8), Dense (512, ReLU), another Dropout (0.8), and Dense (15, softmax) layers for custom classification.
- Compilation and Training. Compiled with Adam optimizer (learning rate 0.0001) and categorical crossentropy loss. Includes ReduceLROnPlateau and EarlyStopping callbacks, trained for up to 300 epochs with early stopping based on validation loss.





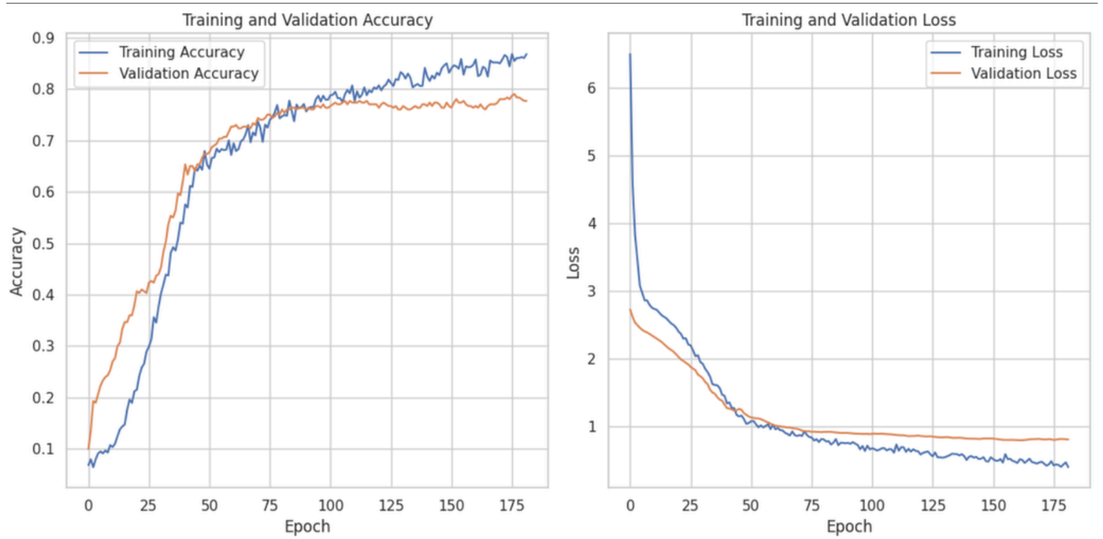






Evaluation

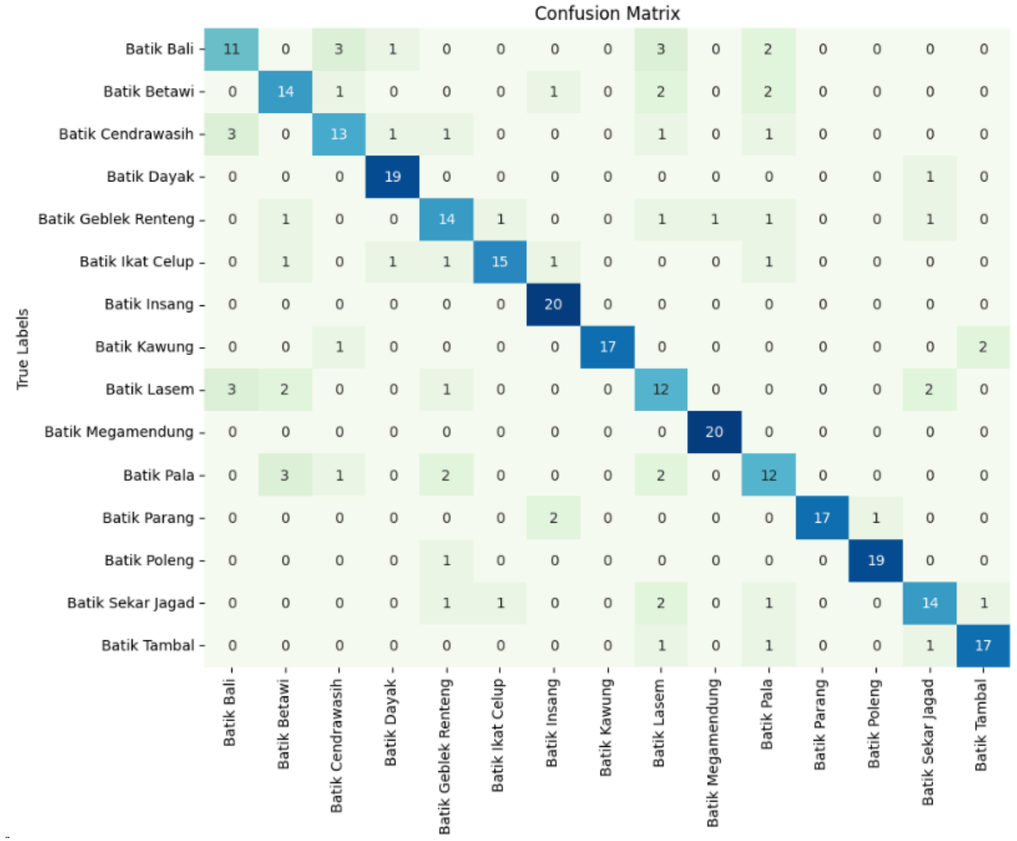
Plotting:

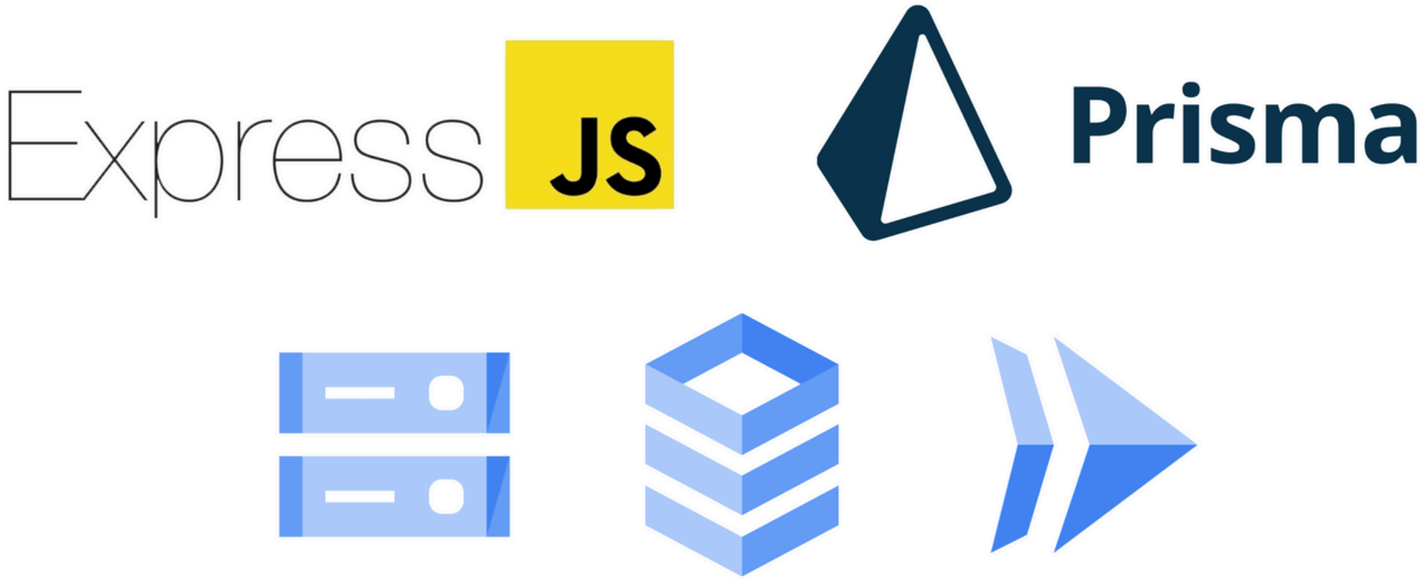


Final_Res

```
loss: 0.3971 - accuracy: 0.8676 - val_loss: 0.8057 - val_accuracy: 0.7767 - lr: 1.0000
```

	precision	recall	f1-score	support
Batik Bali	64.705882	55.0	59.459459	20.00
Batik Betawi	66.666667	70.0	68.292683	20.00
Batik Cendrawasih	68.421053	65.0	66.666667	20.00
Batik Dayak	86.363636	95.0	90.476190	20.00
Batik Geblek Renteng	66.666667	70.0	68.292683	20.00
Batik Ikat Celup	88.235294	75.0	81.081081	20.00
Batik Insang	83.333333	100.0	90.909091	20.00
Batik Kawung	100.000000	85.0	91.891892	20.00
Batik Lasem	50.000000	60.0	54.545455	20.00
Batik Megamendung	95.238095	100.0	97.560976	20.00
Batik Pala	57.142857	60.0	58.536585	20.00
Batik Parang	100.000000	85.0	91.891892	20.00
Batik Poleng	95.000000	95.0	95.000000	20.00
Batik Sekar Jagad	73.684211	70.0	71.794872	20.00
Batik Tambal	85.000000	85.0	85.000000	20.00
accuracy	78.000000	78.0	78.000000	0.78
macro avg	78.697180	78.0	78.093302	300.00
weighted avg	78.697180	78.0	78.093302	300.00





Cloud Computing

The backend API was developed using Express.js and Prisma ORM to connect it to the database. For cloud services, it uses Cloud Storage Bucket to store machine learning models as well as images of each batik motif, Cloud SQL as the database, and Cloud Run to deploy the API so that it can be consumed by the mobile development team.









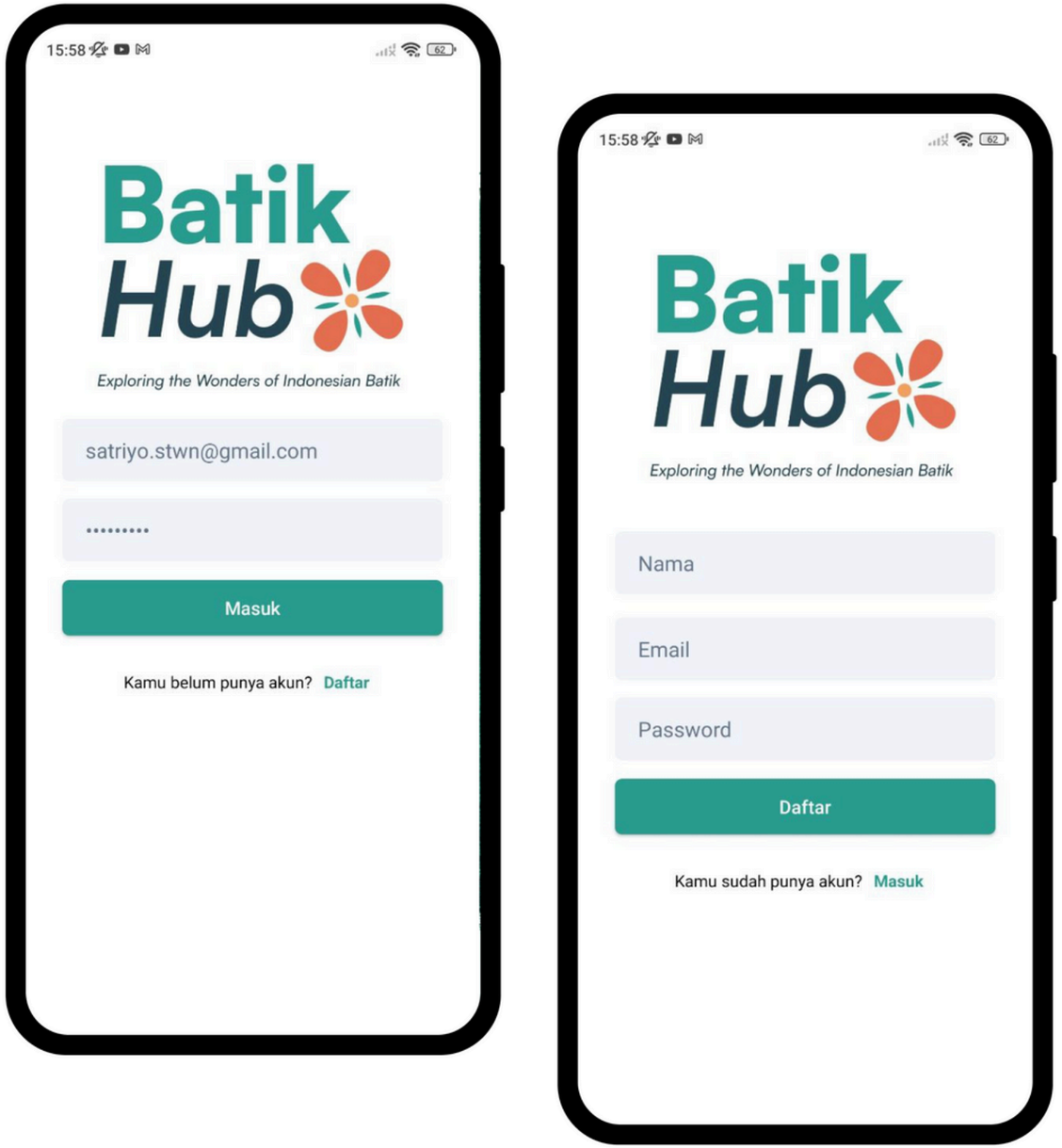


Mobile Development

The mobile application connects in real-time using Firebase Authentication and Firebase Realtime Database, allowing users to view their search history. It also integrates APIs provided by cloud computing to display a list of all batik, search batik by name, and get predictions of batik patterns.

Login & Register Page

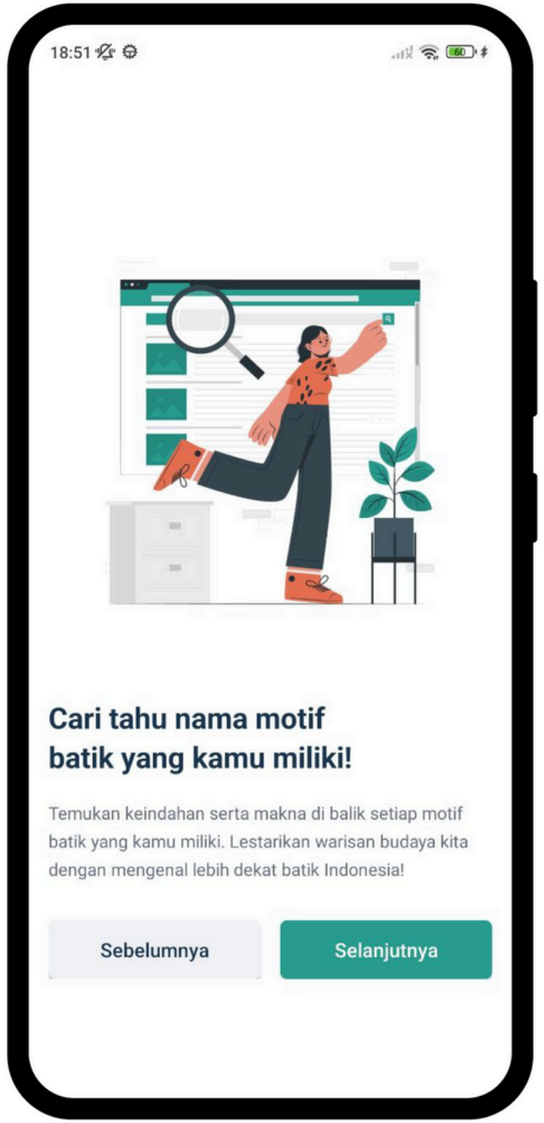
User can log in using the BatikHub account that they already registered.



Information Page

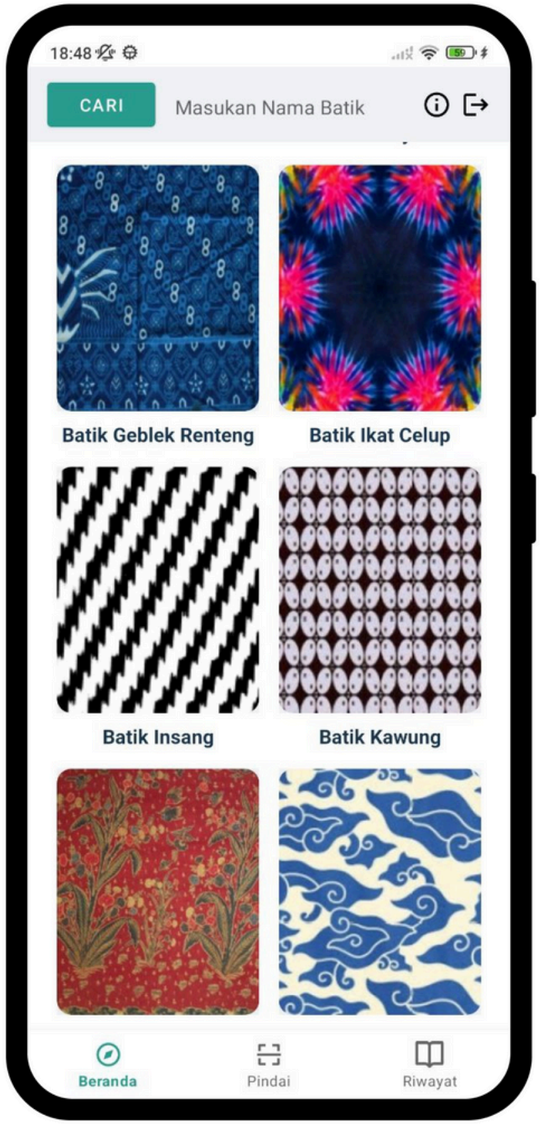
Contains an explanation of the features available in the BatikHub application.

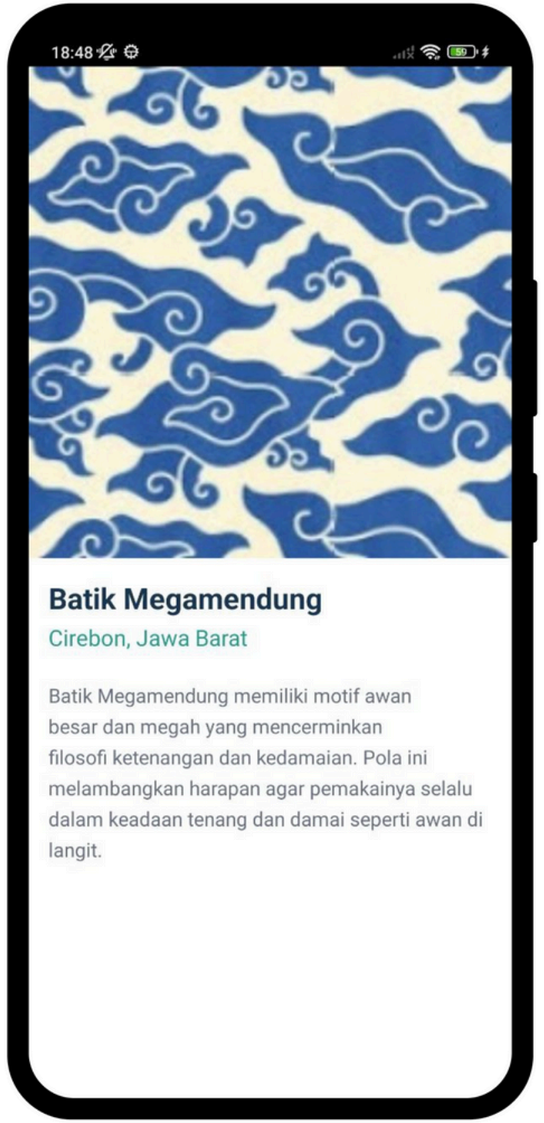




Home Page

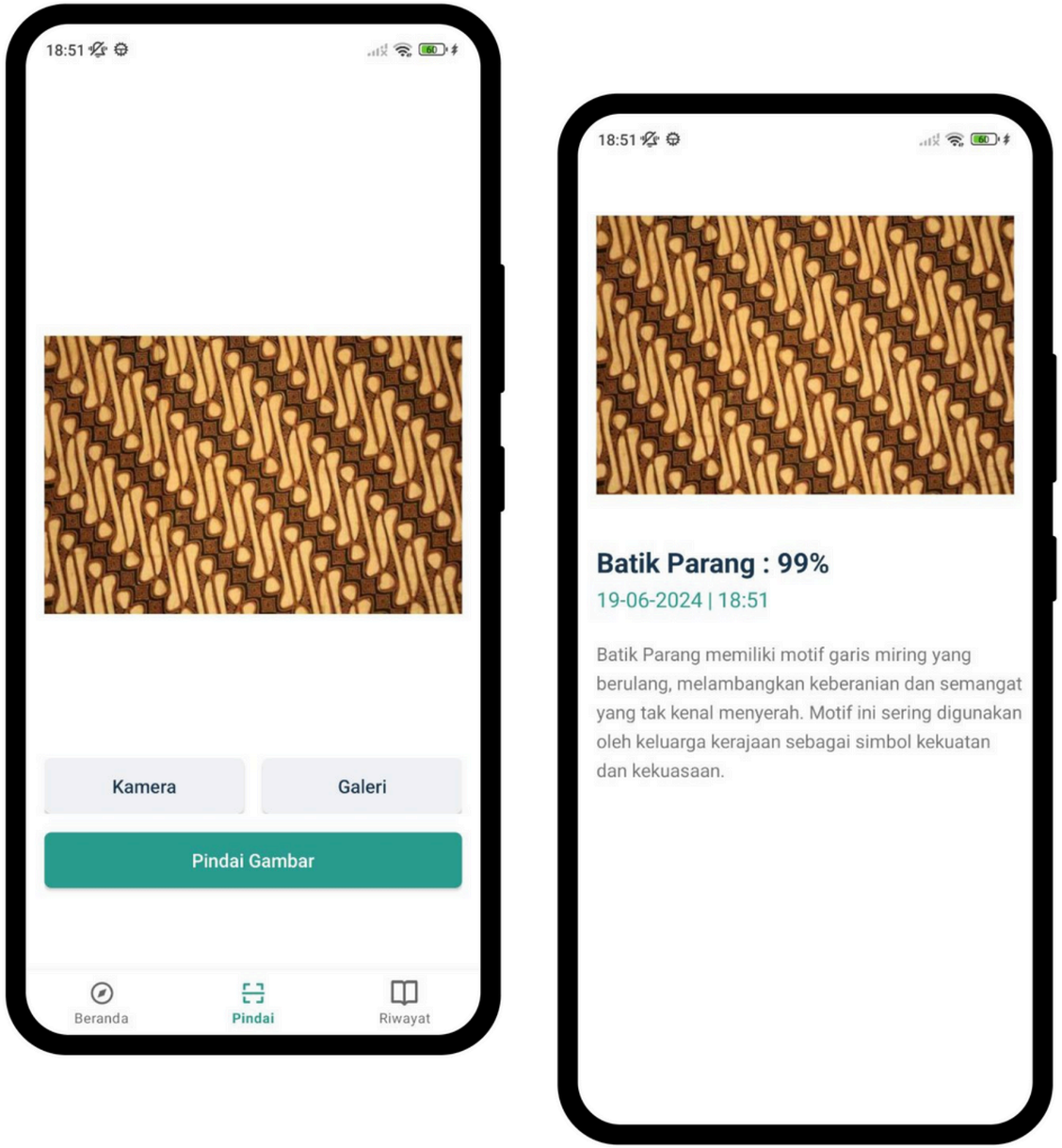
The homepage displays a list of batik with features for logout, batik search, and fetching batik data via API.





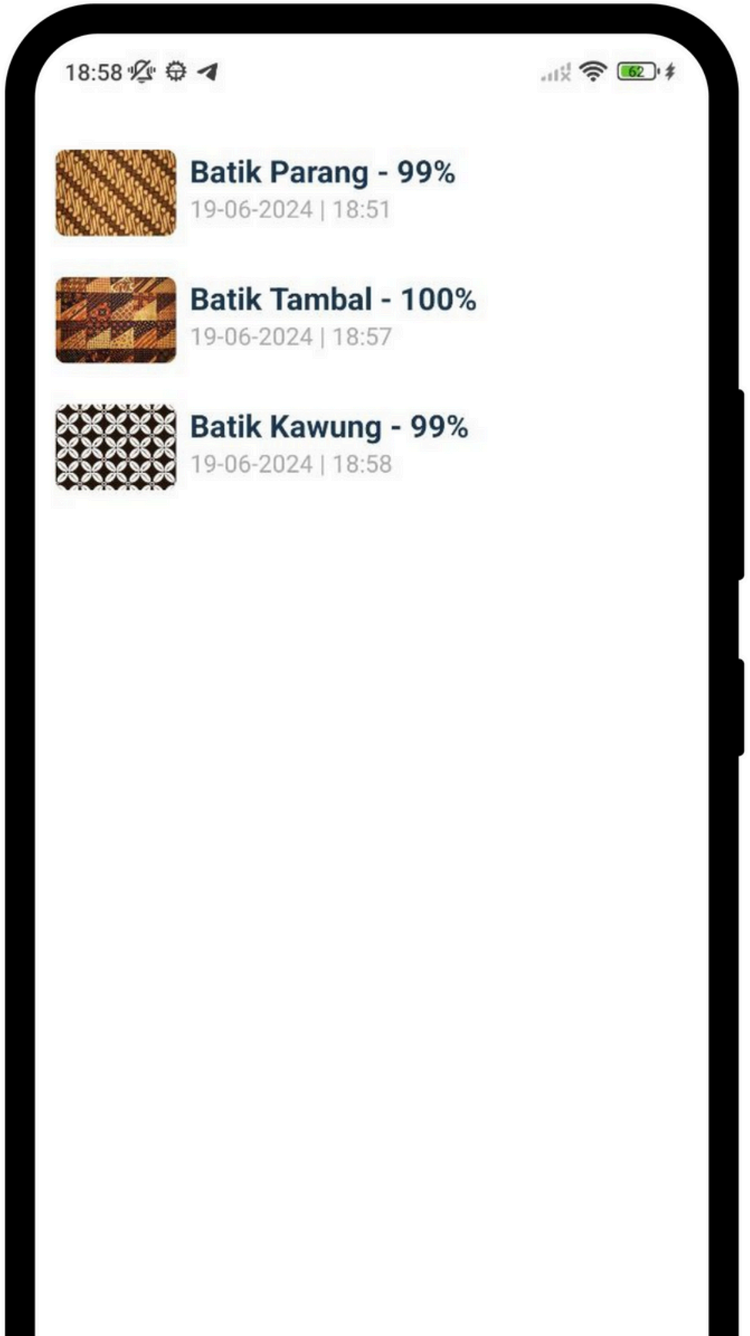
Scan Page

Users can take pictures from the camera or gallery and perform batik image predictions via API. The prediction results are displayed in PredictionResultActivity.



History Page

Users can view the history of batik motif searches displayed on the Prediction Result page.



Target Market



Young Generations

Teens and young adults who frequently use smartphones and mobile apps



Educators and Students

Teachers and students who wants to gain deeper insights into Indonesian batik.

BatikHub's Plan in 2024

Increasing the accuracy of predicting batik motifs using image scan features and Improving and optimizing the motif identification algorithm.



Improve Prediction Accuracy:

- Develop and enhance machine learning models for better motif identification.
- Conduct rigorous testing on new models to ensure reliability.



Implement Enhanced Models:

- Deploy the improved machine learning models into the application.
- Ensure seamless integration and functionality within the existing system.



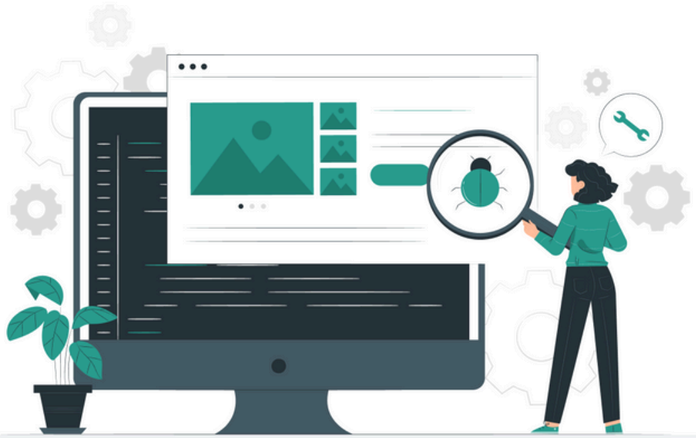
User Feedback Integration:

- Collect and analyze user feedback to identify areas for improvement.
- Iteratively refine models based on user input to continually enhance performance.

Plan in Q3 2024

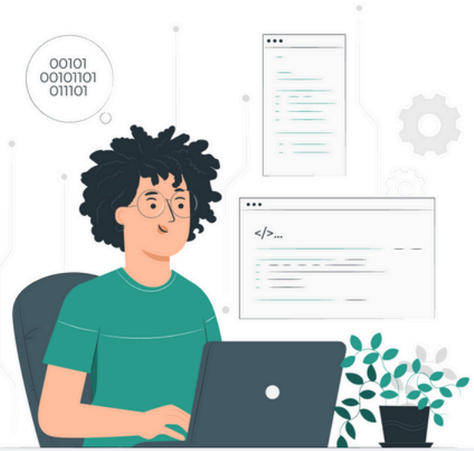
BatikHub's Plan in 2024

Add notification and social media sharing features to enhance user interaction and allow sharing of motif identification results.



Optimize Performance and Bug Fixes:

- Enhance motif identification feature and overall app performance.
- Conduct thorough bug fixes and performance improvements.



Feature Development and Testing:

- Develop notification and social media sharing features.
- Test and validate the accuracy of batik motif predictions.



Official Launch and User Engagement:

- Officially launch the improved motif identification feature.
- Integrate user feedback and enable sharing on social media platforms

Plan in Q4 2024

Budget Plan

No	Description	Budget
1	Team Salary	Rp35.000.000
2	Infrastructure	Rp30.000.000
3	Marketing & Promotions	Rp25.000.000
4	Research & Development	Rp20.000.000
	Total	**Rp110.000.000**

thank you.